

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$$

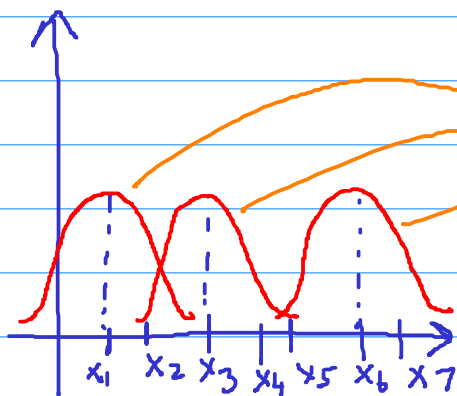
$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \times n \times \sigma^2 = \sigma^2/n$$

$$E(\bar{X}) = \frac{1}{n} \times n \mu = \mu$$

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

$$\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right) \sim N(0, 1)$$



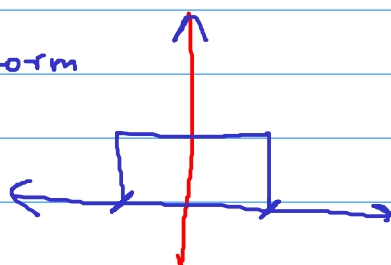
Kernel density estimator
 h
 $=$ bandwidth / smoothing parameter of the KDE

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right) \quad h \text{ small} \rightarrow$$

$$= \frac{1}{n} \sum_{i=1}^n \frac{e^{-\frac{(x - x_i)^2}{2h^2}}}{h\sqrt{2\pi}} \quad \checkmark$$

Uniform

Эпанечников



Kernel ① smooth, diff. function

② symmetric

③ $\lim_{x \rightarrow -\infty} K(x) = 0 = \lim_{x \rightarrow +\infty} K(x)$

Bias of the KDE

point at which the PDF is evaluated

samples

$$E(\hat{p}_n(x_0)) = E\left[\frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - x_0}{h}\right)\right]$$

$= \frac{1}{nh} \times n E\left(K\left(\frac{x_i - x_0}{h}\right)\right) \because \text{all } X_i\text{'s are i.i.d. distr.}$

$$= \frac{1}{h} \int K\left(\frac{x - x_0}{h}\right) p(x) dx$$

$$y = \frac{x - x_0}{h}$$

$$= \frac{1}{h} \int K(y) p(hy + x_0) h dy \quad dy = dx/h$$

$$p(x_0 + hy) = p(x_0) + hy p'(x_0) + \frac{1}{2} h^2 y^2 p''(x_0) + o(h^2)$$

for small h